

Jinuk Moon

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Seoul National University

PROFILE

PhD candidate developing [machine learning interatomic potentials](#) for accelerating catalyst discovery. Developed [CatBench](#), a systematic benchmarking framework that enables reliable evaluation of MLIPs in heterogeneous catalysis. Research focuses on [combining machine learning with first-principles calculations](#) to discover efficient catalytic materials for clean energy applications.

EDUCATION

- **Seoul National University** 2023 – Present
Ph.D. Materials Science and Engineering Seoul, South Korea
 - Machine learning-accelerated computational catalysis for electrocatalyst discovery and design
- **Pohang University of Science and Technology (POSTECH)** 2019 – 2022
B.S. Chemical Engineering Pohang, South Korea
 - Thesis: Study of Metal-Nitrogen-doped carbon catalysts for Chlorine Evolution Reaction combining DFT calculation with Machine Learning
 - GPA: 3.98/4.3 (Summa Cum Laude)
- **KU Leuven** Spring 2022
Exchange Student - Faculty of Science Leuven, Belgium

OPEN SOURCE & CODE

- **CatBench - Benchmark Framework for Machine Learning Interatomic Potentials in Catalysis**
Standardized framework for fair comparison of MLIPs in heterogeneous catalysis
 - pip-installable Python package | catbench.org | github.com/JinukMoon/CatBench

HONORS & AWARDS

- **Presidential Science Scholarship (PhD Track)** 2025 – Present
Ministry of Science and ICT South Korea
- **Best Oral Award** Spring 2025
The Korean Institute of Metals and Materials (KIM) South Korea
 - CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis
- **Best Oral Award** Fall 2024
The Korean Institute of Metals and Materials (KIM) South Korea
 - Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential
- **Best Oral Award** Spring 2024
The Korean Ceramic Society (KCerS) South Korea
 - Selective control of oxygen evolution mechanisms by doping concentrations in RuO₂
- **Best Poster Award** 2025
The 8th International Conference on Advanced Electromaterials (ICAE 2025) South Korea
 - Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution

SKILLS

- **Programming & Frameworks:** Python (Advanced), PyTorch, C/C++ (Intermediate), Bash scripting (Intermediate), HTML
- **Machine Learning:** MLIP Training & Benchmarking (15+ architectures), Dataset Generation, Bayesian Optimization
 - *MLIP Fine-tuning:* Generated 100,000+ in-house DFT calculations, fine-tuned GemNet-OC reducing adsorption energy MAE from 0.48 eV to 0.05 eV (evaluated by CatBench)
 - *Large-scale Molecular Dynamics:* MLIP-accelerated MD simulations (100+ ns) for metal dispersion and structural evolution analysis
 - *High-throughput Screening:* MLIP-accelerated screening of alloy catalysts using bulk structures from Materials Project, OQMD, GNoMe, and OMat datasets
 - *Genetic Algorithm:* Ensemble-GA development with sub-population partitioning and mixing strategies for enhanced global optimization
- **Computational Chemistry:** Density Functional Theory (VASP), Machine Learning Interatomic Potentials (15+ models: UMA, EquiformerV2, GemNet-OC, MACE, etc.), Molecular Dynamics, Electronic Structure Analysis
- **High-Performance Computing:** Supercomputer Cluster Server Management, SLURM, PBS, Batch Job Optimization
- **Research Expertise:** Heterogeneous Catalysis, Single Atom Catalysis, Electrocatalysis (OER, HER, CER), Multiscale Simulation using MLIP, MLIP Benchmarking, ML Analysis of Experimental Characterization
- **Language:** Korean (Native), English (Fluent)

PUBLICATIONS (11 PUBLISHED, 2 SUBMITTED, 7 FIRST-AUTHORED)

[†] Equal contribution, * Corresponding author

- [submitted] Jun, H.[†], Kwon, D.[†], **Moon, J.[†]**, Han, S.[†], Kim, S., Yi, S. Y., Kim, Y., Kang, E., Choi, C. H.* , Han, J. W.* , Kim, W.* , Lee, J.* (2026). [Synergistic Effect of Tuning Reaction Mechanism and Interfacial Water for Efficient Acidic Water Oxidation](#). submitted.
- [submitted] Jo, H.[†], Sohn, Y.[†], **Moon, J.[†]**, Han, J. W.* , Kim, W.* , Yong, K.* (2026). [Synthesis-stage Interfacial Engineering of Edge-rich \$\beta\$ -Co\(OH\)₂ for Efficient Nitrate to Ammonia Electrosynthesis](#). submitted.
- [1] Kim, J.[†], **Moon, J.[†]**, Cho, K.[†], Kim, J.[†], Han, M. H., Kim, S., Han, J., Choi, D., Jun, H., Jang, J. H., Oh, H.-S., Cho, S. K.* , Kim, W.* , Han, J. W.* , Lee, J.* (2026). [Size-Dependent Dynamic Surface States of Ru Clusters: Modulating Interfacial Water Structure for Enhanced Alkaline Hydrogen Evolution Reaction](#). *Energy & Environmental Science*.
- [2] Kim, J.[†], Yang, G. G.[†], Lee, W.[†], Nam, S., **Moon, J.**, Yang, H., Seo, J., Choi, Y., Jeong, B.* , Han, J. W.* , Kim, S. O.* , Jung, W.* (2026). [Quantitative Electrochemical Evaluation of Metal Nanocatalysts for Oxygen Exchange on Solid Oxide Cell Electrodes](#). *Energy & Environmental Science*.
- [3] Lim, H.[†], Choung, S.[†], **Moon, J.**, Han, J. W.* (2026). [Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration](#). *npj Computational Materials*.
- [4] Park, K.[†], Lee, W.[†], Park, J., Jo, M., Bae, Y., Kim, M., Shin, S., **Moon, J.**, O'Hayre, R.* , Han, J. W.* , Song, S.* , Park, J. Y.* (2026). [Composite air electrodes based on BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3- \$\delta\$} for reversible solid oxide cells](#). *Nature Energy*.
- [5] Choung, S.[†], Kim, M.[†], **Moon, J.[†]**, Han, J. W.* (2025). [From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials](#). *ACS Energy Letters*, 10, 6288-6296.
- [6] **Moon, J.**, Jeon, U., Choung, S., Han, J. W.* (2025). [CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis](#). *Cell Reports Physical Science*, 6, 102968.

- [7] Jung, H.[†], Lee, Y.[†], Cho, K.[†], Kim, T. Y., Song, J., Jung, H. S., Park, S. I., Lee, Y., **Moon, J.**, Park, W., Nam, J., Park, S., Kim, W., Han, J. W.* (2025). [Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction Under Seawater](#). *Advanced Functional Materials*, e16387.
- [8] Jun, H.[†], Kang, E.[†], **Moon, J.**[†], Kim, H., Han, S., Choung, S., Kim, S., Yi, S. Y., Kang, E., Choi, C. H.* , Han, J. W.* , Lee, J.* (2025). [Quantity effect of heteroatom incorporation on the oxygen evolution mechanism in ruthenium oxide](#). *Chem*, 11(5), 102367.
- [9] Choung, S.[†], Park, W.[†], **Moon, J.**[†], Han, J. W.* (2024). [Rise of machine learning potentials in heterogeneous catalysis: Developments, applications, and prospects](#). *Chemical Engineering Journal*, 494, 152757.
- [10] Choung, S.[†], Yang, H.[†], **Moon, J.**, Park, W., June, H., Lim, C., Han, J. W.* (2024). [Theoretical tuning of local coordination environment of metal-nitrogen doped carbon catalysts for selective chlorine-evolution reaction](#). *Catalysis Today*, 425, 114358.
- [11] Choung, S., Kim, Y., **Moon, J.**, Roh, J., Hwang, J., Han, J. W.* (2023). [Unveiling the catalyst deactivation mechanism in the non-oxidative dehydrogenation of light alkanes on Rh\(111\): Density functional theory and kinetic Monte Carlo study](#). *Catalysis Today*, 411-412, 113819.

CONFERENCE PRESENTATIONS

- [1] **Moon, J.** (2025). [CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis](#) (Poster). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [2] **Moon, J.** (2025). [Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Graph Neural Network](#) (Oral Presentation). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [3] **Moon, J.** (2026). [Synergistic Effect of Oxygen Evolution Mechanism and Interfacial Water Dynamics Revealed by MLIP Distillation](#) (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.
- [4] **Moon, J.** (2026). [Machine Learning Accelerated Large Scale Analysis of Interfacial Water Structure on Size-controlled Ru Clusters for Alkaline Hydrogen Evolution](#) (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.
- [5] **Moon, J.** (2025). [Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution](#) (Poster). *The 8th International Conference on Advanced Electromaterials (ICAE 2025)*, Jeju, South Korea.
- [6] **Moon, J.** (2025). [Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution](#) (Poster). *Korean Institute of Metals and Materials (KIM)*, Gwangju, South Korea.
- [7] **Moon, J.** (2025). [Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Fine-tuned Machine Learning Interatomic Potential](#) (Oral Presentation). *Korean Institute of Chemical Engineers (KICHE)*, Jeju, South Korea.
- [8] **Moon, J.** (2025). [CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis](#) (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.
- [9] **Moon, J.** (2025). [CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis](#) (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.
- [10] **Moon, J.** (2024). [Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential](#) (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, PyeongChang, South Korea.

- [11] **Moon, J.** (2024). [Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential](#) (Oral Presentation). *23rd Korea Japan Students' Symposium*, Seoul, South Korea.
- [12] **Moon, J.** (2024). [Selective Control of Oxygen Evolution Reaction Mechanisms by Adjusting Mn Doping Concentrations in RuO₂](#) (Oral Presentation). *The Korean Ceramic Society (KCerS)*, Busan, South Korea.